5. Illustrate the physical and probability-based Models used in Compression (lossless), highlighting their application

A. Lossless compression reduces file size without losing information. The original data can be perfectly reconstructed. Based on modeling technique, it is classified into:

1. Physical Models
2. Probability-Based Models

1. Physical Models.

physical models exploit the structural repetition of patterns in data. They remove spatial redundancy by replacing repeated sequences with shorter references.

Working Principle.

- Detect repeated strings
- Replace them with pointer (length or dictionary reference.
- Reconstruct original data during decoding.

Techniques.

a) Run Length Encoding (RLE)

Store repeated symbols as

Symbol + count.

Example : AAAA → A 4

Used in bitmap images and fax transmission

- Lempel-Ziv (LZ77/LZ78)

Uses sliding window or dictionary.
Repeated patterns are encoded as (offset, length).
Used in ZIP and GZIP.

Advantages: Simple, fast
Limitation: Inefficient if repetition is low.

- Probability-Based Models.

→ These models use statistical frequency of symbols. Frequently occurring symbols are assigned shorter codes, reducing statistical redundancy.

Working Principle.

- Calculate symbol probability.
- Assign variable length codes.
- Minimize average code length.

Techniques.

a) Huffman Coding
- Constructs binary tree using frequencies
- Generates prefix-free codes.
- Used in image and audio compression.

b) Arithmetic Coding
- Represents entire message as a fraction between 0 and 1
- provides higher compression efficiency.

Advantages: Better compression ratios
Limitation: More complex.

Conclusion.
Physical Models remove redundancy using repeated patterns, while probability-based models use symbol frequency to assign optimal codes.